INCH-POUND

MIL-DTL-20549C
10 August 2024
SUPERSEDING
MIL-L-20549B
12 January 1983

DETAIL SPECIFICATION

LEAD NITRATE, TECHNICAL

Reinstated after 10 August 2024 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of lead nitrate.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at (https://quicksearch.dla.mil/.)

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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2.3 Non-Government standards and publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1193	-	Standard Specification for Reagent Water
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals

(Copies of these documents are available from https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Chemical requirements. Lead nitrate shall conform to the chemical characteristics of Table I when tested as specified therein.

TABLE I. Chemical requirements.

Characteristic	Requirement (weight percent, wt. %)
Assay (as $Pb(NO_3)_2$)	98.0 minimum (min.)
Water content	1.25 maximum (max.)
Water-insoluble matter	0.20 max.
Acidity (as HNO_3)	0.50 max.
Copper	0.002 max.

3.3 Workmanship. Lead nitrate shall be white in color, uniform in quality and free from visible impurities and foreign material such as dirt, rust, paint, wooden particles or any other foreign objects.

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4. VERIFICATION

TABLE II. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X		X		4.2
3.1.2	Conformance	X		X			X	4.3
3.2	Assay (as $Pb(NO_3)_2$)	X				X	X	4.4.1
3.2	Water content	X				X	X	4.4.2
3.2	Water-insoluble matter	X				X	X	4.4.3
3.2	Acidity (as HNO_3)	X				X	X	4.4.4
3.2	Copper	X				X	X	4.4.5
3.3	Workmanship			X		X	X	4.4.6

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).
b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table III. Samples collected in accordance with 4.3.2 shall be submitted for examination. The sample shall be taken from a batch or lot representative in size to normal production or two hundred pounds, whichever is greater. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

TABLE III. First article acceptance tests.

Category	Examination or test	Requirement paragraph	Inspection method
Critical	None defined		
Major			
101	Assay (as $Pb(NO_3)_2$)	3.2	4.4.1
102	Water content	3.2	4.4.2
103	Water-insoluble matter	3.2	4.4.3
104	Acidity (as HNO_3)	3.2	4.4.4
105	Copper	3.2	4.4.5
Minor			
201	Workmanship	3.3	4.4.6

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4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916 paragraph "Formation and identification of lots or batches." Each batch shall consist of that quantity of lead nitrate that has been subjected to the same chemical and physical mixing process intended to make the final product homogenous. A lot shall consist of one or more acceptable batches of lead nitrate produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions. A lot shall consist of the lead nitrate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.3.2 Lead nitrate sampling. Sampling for inspection shall be performed at either the supplier location where the lead nitrate is packed, or at the manufacturer's receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

4.3.2.1 Sampling at the supplier. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. There shall be no composite sampling. All samples shall be verified in accordance with 4.4. If any sample fails to comply with any applicable requirements the batch shall be rejected.

4.3.2.2 Sampling at the receiving facility. Lead nitrate packaged in shipping containers (e.g., drums, boxes, etc.) shall be sampled using Verification Level II of MIL-STD-1916. Each shipping container shall be sampled in three locations in accordance with ASTM E300 using a thief stick. The representative specimen from each shipping container shall be not less than 200 g and shall be placed in a suitable clean, dry container labeled to identify the lot and shipping container from which it was taken. There shall be no composite sampling. There shall be no reduced sampling. Every lot received shall be tested in accordance with 4.4. If any sample fails to meet any test requirement, the lot represented by the sample shall be rejected.

4.3.3 Classification of characteristics.

4.3.3.1 Lead nitrate.

Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Assay (as $Pb(NO_3)_2$)	4.3.2	3.2	4.4.1
102	Water content	4.3.2	3.2	4.4.2
103	Water-insoluble matter	4.3.2	3.2	4.4.3
104	Acidity (as HNO_3)	4.3.2	3.2	4.4.4
105	Copper	4.3.2	3.2	4.4.5

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4.3.3.1 Lead nitrate. - Continued				
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Minor				
201	Workmanship	4.3.2	3.3	4.4.6

4.4 Test methods and procedures. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.4.1 Assay. Weigh to the nearest milligram (mg) approximately 1 g of the specimen into a 250 milliliter (mL) beaker and dissolve in 100 mL of water. Add drop by drop, with stirring, 10 mL of saturated sodium sulfate solution containing 5 percent sulfuric acid and allow to stand for at least 2 hours. Filter through a tared fritted glass crucible of medium porosity and wash with 1 percent alcohol (ethanol or methanol are acceptable). Heat the crucible and precipitate in a muffle furnace at 200º ± 10ºC to constant weight, cool in a desiccator, and weigh. Calculate the percent by weight lead nitrate as follows:

$$\text{Percent by weight lead nitrate} = \frac{109.2 \times (A - B)}{C}$$

where: A = Weight of crucible and precipitate in grams,
B = Weight of crucible in grams, and
C = Weight of specimen in grams.

4.4.2 Water content. Weigh to the nearest milligram approximately 10 g of the specimen into a tared weighing bottle. Ensure there are no clumps or agglomerates in the sample. Heat the uncovered bottle and contents at 125ºC ± 2ºC for 5 hours, cool in a desiccator, and weigh. Calculate the percent by weight water content as follows:

$$\text{Percent by weight water content} = \frac{100 \times (D - E)}{D - F}$$

where: D = Weight of bottle and specimen before heating in grams,
E = Weight of bottle and specimen after heating in grams, and
F = Weight of tared weighing bottle.

4.4.3 Water-insoluble matter. Weigh to the nearest milligram approximately 25 g of the specimen in a 500 mL beaker. Add 250 mL of water and heat until the specimen is completely dissolved. Filter the hot solution through a tared filter crucible and wash thoroughly with hot water. Dry the crucible and residue at 110ºC ± 2ºC for 2 hours, cool in a desiccator, and weigh. Calculate the percent weight water-insoluble matter as follows:

$$\text{Percent by weight water-insoluble matter} = \frac{100 \times (G - H)}{J}$$

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where: G = Weight of crucible and residue in grams,
H = Weight of crucible in grams, and
J = Weight of specimen used in grams.

4.4.4 Acidity. Determine acidity content in accordance with the following procedure.

4.4.4.1 Equipment and supplies.

a. pH-meter with temperature compensation
b. pH combination electrode, double-junction, conditioned prior to use per manufacturer instructions,
c. Manual or automated means of stirring,
d. Burette 50 mL, Class A.
e. Alternately, an automatic titrator may be used with potentiometric endpoint detection may be used.

4.4.4.2 Reagents.

a. Deionized or distilled water
b. Sodium hydroxide, 0.05 N , standardized against reference-grade potassium hydrogen phthalate to a faint-pink phenolphthalein endpoint. Alternately, a certified, commercial solution may be used.
c. pH 4 and pH 7 calibration buffers, NIST-traceable

4.4.4.3 Procedure. Calibrate the pH meter using the buffer solutions. Weigh 25.0 ± 0.1 g of the specimen into a 600 mL beaker, dissolve in 500-mL of water, and titrate with 0.05N sodium hydroxide. The endpoint, which occurs at pH 4.0-4.3, shall be detected potentiometrically. Calculate the percent acidity as nitric acid as follows:

$$\text{Percent by weight acidity (as } HNO_3) = \frac{6.301 \times K \times L}{M}$$

where: K = Milliliters of sodium hydroxide solution used,
L = Normality of sodium hydroxide solution used, and
M = Weight of specimen in grams

4.4.5 Copper. Weigh 1.000 g of the specimen and dissolve in 15 mL of water. Add 5 mL of 10 percent ammonium acetate solution, 0.5 mL of ammonium thiocyanate solution (300 g of ammonium thiocyanate per liter), 2 mL of acetic acid, 0.5 mL of pyridine, and 10 mL of chloroform. Similarly prepare a standard for comparison by weighing and dissolving 0.02 mg of copper ion (Cu), then adding 5 mL of 10 percent ammonium acetate solution, 0.5 mL of ammonium thiocyanate solution, 2 mL of acetic acid, 0.5 mL of pyridine, and 10 mL of chloroform. Shake both solutions vigorously and allow to separate. Any greenish yellow color produced in the chloroform (lower) layer of the sample shall be not darker than that produced in the standard.

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4.4.6 Workmanship. Take approximately 100 grams of the material and spread the composition out on a clean white surface. The sample shall be white in color, uniform and free from visible impurities, foreign matter or other foreign objects. Failure to meet the workmanship inspection or during following production processes shall result in batch or lot rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Lead nitrate is intended for use in the manufacture of lead styphnate and lead azide.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Provisions for submission/resubmission of first article samples.
c. Quantity required and delivery schedules.
d. Packaging requirements (see 5.1).
e. Requirements for submission of acceptance inspection equipment.
f. Certificate(s) of Analysis reporting the test results for the batch(es) or container(s) tested.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process of subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially hazardous to personnel. Lead nitrate is a strong oxidizer, toxic by ingestion and inhalation. All applicable safety rules, regulations, and procedures should be followed in the sampling and testing of this material.

6.5 Technical agency. US Army Combat Capability Development Command - Armaments Center (DEVCOM AC) is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be

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submitted through the contracting officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.6 Approval of equivalent test methods. Prior approval by the Contracting Officer should be required for use of equivalent test methods and/or instruments. A description of the proposed method and/or instrumentation should be submitted through the Contracting Officer to the Technical Agency (see 6.5). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.7 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency (See 6.5) responsible for preparation of this specification.

6.8 Incorporated changes. This specification was updated from MIL-A-20549B to MIL-DTL-20549C as part of the Munitions Chemical Specification Modernization (MCSM) program. All test procedures were verified and updated as needed to produce valid results. Reference documents have been verified, updated, and hyperlinked to sources. Sampling procedures have been updated.

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6.9 Subject term (key word) listing.

Precursor
Lead styphnate
Lead azide
MCSM

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – OS
DLA – GS

Preparing activity:
Army – AR
Project No. 6810-2023-014

Review activities:
Army – EA, MD

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